Construction of DNA Codes with R-Constraint, RC-Constraint, and Fixed GC Content

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Basics of DNA

What is DNA?

DNA (Deoxyribonucleic Acid) is the natural carrier of genetic information in living organisms.

DNA sequences are formed using four nucleotides:

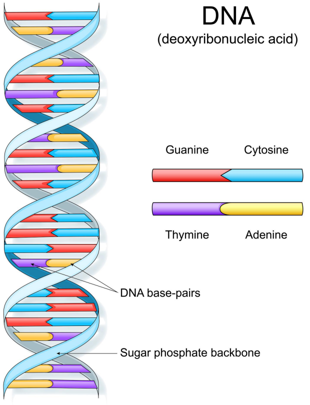
A	T	G	C
Adenine	Thymine	Guanine	Cytosine

Watson–Crick Complementarity

DNA bases pair according to

$$A \leftrightarrow T, \qquad G \leftrightarrow C$$

This complementary property plays an important role in DNA storage and DNA computing.



Motivation for DNA Codes

- **DNA Computing**
  In 1994, Leonard Adleman[1] introduced DNA computing by solving a small Hamiltonian path problem using DNA strands.

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  Researchers have studied DNA codes with reverse, reverse-complement, GC-content, and distance constraints to avoid hybridization and secondary structures.[3]

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  These developments established DNA as a promising medium for
  - Large-scale information storage.
  - Molecular computation.

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Why Are Constraints Needed?

Practical Challenges in DNA Storage

- Sequencing errors
- Unwanted hybridization
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Coding-Theoretic Solution

DNA codes are designed to satisfy biological constraints in order to improve

- reliability,
- stability,
- error correction capability.

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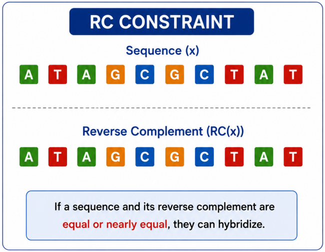

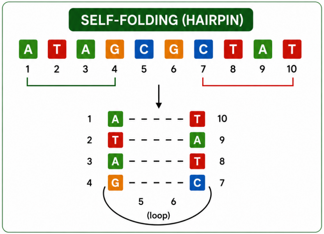

Codes

Codes over finite fields

- Symbols are taken from a finite field $\mathbb{F}_q$.
- A code is a subset $\mathcal{C} \subseteq \mathbb{F}_q^n$.
- For $\mathbf{u} = (u_1, u_2, \ldots, u_n), \mathbf{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{F}_q^n$,
  - The Hamming distance between **u** and **v**,

$$d_H(\mathbf{u},\ \mathbf{v}) = |\{i \mid u_i \neq v_i\}|.$$

  - The minimum Hamming distance of a code $\mathcal{C} \subseteq \mathbb{F}_q^n$,

$$d_H(\mathcal{C}) = \min\{d_H(\mathbf{c},\ \mathbf{c}') \mid \mathbf{c}, \mathbf{c}' \in \mathcal{C},\ \mathbf{c} \neq \mathbf{c}'\}.$$

Linear codes

- $\mathcal{C}$ is a vector subspace of $\mathbb{F}_q^n$.
- Parameters: $[n, k, d]_q$ (length, dimension, distance).

DNA Codes as Codes over $\mathbb{F}_4$

DNA Alphabet

$$\mathscr{A} = \{A, C, G, T\}$$

Watson–Crick Complement

$$\hat{A} = T, \quad \hat{T} = A, \quad \hat{G} = C, \quad \hat{C} = G.$$

A non-empty subset of $\mathscr{A}^n$ is called a DNA code of length n.

For

$$\mathbf{a} = (a_0, a_1, \ldots, a_{n-1}) \in \mathscr{A}^n,$$

the reverse and reverse-complement are

$$\mathbf{a}^R = (a_{n-1}, \ldots, a_0),$$

$$\mathbf{a}^{RC} = (\hat{a}_{n-1}, \ldots, \hat{a}_0).$$

GC-weight of a : No. of components belonging to $\{G, C\}$.

Connection with $\mathbb{F}_4$:

Identify

$$\mathscr{A} \longleftrightarrow \mathbb{F}_4 = \{0, \alpha, \alpha+1, 1\},$$

where $\alpha^2 + \alpha + 1 = 0$.

A	C	G	T
0	α	$\alpha+1$	1

Hence DNA codes can be viewed as codes over $\mathbb{F}_4$.

$$\{0, 1\} \simeq \{A, T\},$$
$$\{\alpha, \alpha+1\} \simeq \{C, G\}.$$

Moreover,

$$\hat{\beta} = \beta + 1, \qquad \forall\, \beta \in \mathbb{F}_4.$$

Constraints on DNA codes[4]

1. **Hamming distance constraint**

   $$d_H(\mathbf{c}, \mathbf{c}') \geq d \quad \forall\, \mathbf{c}, \mathbf{c}' \in \mathcal{C},\; \mathbf{c} \neq \mathbf{c}'.$$

   Ensures reliable error detection and correction.

2. **Reverse constraint (R-constraint)**

   $$d_H(\mathbf{c}^R, \mathbf{c}') \geq d \quad \forall\, \mathbf{c}, \mathbf{c}' \in \mathcal{C}.$$

   Avoids undesirable similarity with reversed DNA strands.

3. **Reverse-complement constraint (RC-constraint)**

   $$d_H(\mathbf{c}^{RC}, \mathbf{c}') \geq d \quad \forall\, \mathbf{c}, \mathbf{c}' \in \mathcal{C}.$$

   Prevents unwanted hybridization between DNA strands.

4. **Fixed GC-content**

   Every codeword in $\mathcal{C}$ has the same GC-weight.

   Maintains thermal stability.

[4] P. Gaborit and O. D. King, *Theoretical Computer Science*, 2005.

Decomposition of Reversible and Reversible-complement Codes

Reversible code:
Let $\mathcal{C}$ be a reversible code over $\mathbb{F}_4$, i.e.,

$$\mathbf{c}^R \in \mathcal{C} \qquad \forall\, \mathbf{c} \in \mathcal{C}.$$

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$$\mathcal{C} = \mathcal{C}' \cup \mathcal{C}'',$$

where

$$\begin{aligned} \mathcal{C}' &= \{\mathbf{c} \in \mathcal{C} \mid \mathbf{c}^R = \mathbf{c}\}, \\ \mathcal{C}'' &= \{\mathbf{c} \in \mathcal{C} \mid \mathbf{c}^R \neq \mathbf{c}\}. \end{aligned}$$

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$$\mathcal{C}'' = \tilde{\mathcal{C}''} \cup \tilde{\mathcal{C}''}^R,$$

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$$|\mathcal{C}| = |\mathcal{C}'| + 2|\tilde{\mathcal{C}''}|.$$

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Construction of the Code $\mathcal{C}_s$

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- For $1 \leq t \leq s-1$, and $0 \leq \ell \leq 2^{s-t}-1$, define

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$$\mathcal{A}_s^t = \left\{ \sum_{\ell \in \Gamma} p_\ell^t(x) \mid \Gamma \subseteq \Lambda_s^t,\ |\Gamma| = 2j \right\}, \quad \text{where } 0 \leq j \leq 2^{s-t-1}.$$

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- $$\mathcal{C}_s = \sum_{t=1}^{s-1} \mathcal{F}_s^t = \{f_1(x) + f_2(x) + \ldots + f_{s-1}(x) \mid f_t(x) \in \mathcal{F}_s^t\}.$$

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- $$g_{s+1}(x) = 1 + x^2 + x^4 + \cdots + x^{2^s-2}.$$
- The set $\mathcal{G} = \{g_j(x) \mid 1 \leq j \leq s+1\}$ forms a basis of $\mathcal{C}_s^{\perp}$.

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$$d(x) \neq d'(x) \implies |P \cap Q| = 2^{s-3}.$$

Properties of $\mathcal{C}_s$ & $\mathcal{C}_s^\perp$

Closure of $\mathcal{C}_s$ Under Reverse

For every $c(x) \in \mathcal{C}_s, \quad c(x)^R \in \mathcal{C}_s$.[5]

[5] S. Bansal, P.K. Kewat, "A family of optimal dual-containing and reversible linear codes over $\mathbb{F}_4$," Finite Fields Their Appl., 2026.

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Closure of $\mathcal{C}_s^{\perp}$ Under Reverse

For $c(x) \in \mathcal{C}_s^{\perp}$

$$c(x) = \sum_{j=1}^{s+1} \beta_j g_j(x), \qquad \beta_j \in \mathbb{F}_4.$$

Since

$$g_1(x)^R = g_1(x) \quad \text{and} \quad g_j(x)^R = g_j(x) + g_1(x), \quad 2 \le j \le s+1.$$

Hence,

$$g_j(x)^R \in \mathcal{C}_s^{\perp} \quad \forall\, 1 \le j \le s+1 \implies c(x)^R \in \mathcal{C}_s^{\perp}.$$

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Hence,

$$g_j(x)^R \in \mathcal{C}_s^{\perp} \quad \forall\, 1 \leq j \leq s+1 \implies c(x)^R \in \mathcal{C}_s^{\perp}.$$

Reverse-Complement Closure of $\mathcal{C}_s$ and $\mathcal{C}_s^{\perp}$

Since $c(x)^{RC} = c(x)^R + g_1(x)$, both $\mathcal{C}_s$ and $\mathcal{C}_s^{\perp}$ are also closed under reverse-complement.

[5] S. Bansal, P.K. Kewat,"A family of optimal dual-containing and reversible linear codes over $\mathbb{F}_4$," Finite Fields Their Appl., 2026.

Classification of Codewords in $\mathcal{C}_s^\perp$

Every codeword of $\mathcal{C}_s^\perp$ can be written as

$$c(x) = a_1 g_1(x) + d(x) + a_{s+1} g_{s+1}(x) + \alpha\big(b_1 g_1(x) + d'(x) + b_{s+1} g_{s+1}(x)\big),$$

where $a_1, b_1, a_{s+1}, b_{s+1} \in \mathbb{F}_2, \quad d(x), d'(x) \in D_s.$

Four Main Cases: The codewords are classified according to the pair $(d(x), d'(x))$.

1. $d(x) = 0,\ d'(x) = 0$
2. $d(x) = 0,\ d'(x) \neq 0$
3. $d(x) \neq 0,\ d'(x) = 0$
4. $d(x) \neq 0,\ d'(x) \neq 0$

For each case, the 16 choices of $(a_1, b_1, a_{s+1}, b_{s+1}) \in \mathbb{F}_2^4$ produce 16 disjoint subcases.

Decomposition of $\mathcal{C}_s^\perp$

$\mathcal{C}_s^\perp =$ Union of 64 mutually disjoint subcodes.

Self-Reversible and Self-Reverse-Complement Elements in D_s

- For $d(x) = \sum_{j=2}^{s} a_j g_j(x) \in D_s$, we have

$$d(x)^R = d(x) + g_1(x) \sum_{j=2}^{s} a_j,$$

$$d(x)^{RC} = d(x) + g_1(x)\left(1 + \sum_{j=2}^{s} a_j\right).$$

  - $wt_H(a_2, \ldots, a_s)$ even $\implies$ $d(x)^R = d(x)$.
  - $wt_H(a_2, \ldots, a_s)$ odd $\implies$ $d(x)^{RC} = d(x)$.

- Among the 2^{s-1} elements of D_s, 2^{s-2} are self-reversible and 2^{s-2} are self reverse-complement.

Analysis of the 64 Subcodes

For each subcode, we determine:

1. Whether the codewords satisfy $c(x)^R = c(x),\ c(x)^{RC} = c(x)$, or neither condition.
2. $(n_R,\ n_{RC},\ n_O)$, where
   - n_R: The no. of self-reversible codewords,
   - n_{RC}: The no. of self reverse-complement codewords,
   - n_O: The no. of remaining codewords.
3. The GC-weight of the codewords.
4. The complete weight enumerator (CWE).

Summing the contributions of all 64 subcodes yields

- The complete weight enumerator of $\mathcal{C}_S^{\perp}$,
- The GC-weight enumerator.

The case $d(x) = 0$, $d'(x) \neq 0$ and $(a_1, b_1, a_{s+1}, b_{s+1}) = (0, 0, 0, 1)$

- $$\begin{aligned} c(x) &= \alpha g_{s+1}(x) + \alpha d'(x), \\ c(x)^R &= \alpha g_{s+1}(x) + \alpha g_1(x) + \alpha d'(x)^R, \\ c(x)^{RC} &= \alpha g_{s+1}(x) + (1+\alpha) g_1(x) + \alpha d'(x)^R. \end{aligned}$$

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- Among the nonzero elements of D_s,
  - $2^{s-2} - 1$ elements satisfy $d'(x)^R = d'(x)$,
  - while the remaining 2^{s-2} elements satisfy $d'(x)^{RC} = d'(x)$.

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- $$(n_R, n_{RC}, n_O) = \left(2^{s-2}, 0, 2^{s-2} - 1\right).$$

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- $$(n_R, n_{RC}, n_O) = \left(2^{s-2}, 0, 2^{s-2} - 1\right).$$

- Moreover, $d'(x) = \sum_{\ell \in Q}(x^{2\ell} + x^{2\ell+1})$ implies
$$c(x) = \alpha \sum_{\ell \in \Lambda_s^1 \setminus Q} x^{2\ell} + \alpha \sum_{\ell \in Q} x^{2\ell+1}.$$
Hence, $GCW(c(x)) = 2^{s-1}$, and the complete weight enumerator is
$$(2^{s-1} - 1) x_0^{2^{s-1}} x_\alpha^{2^{s-1}}.$$

Main Results

From the analysis of all 64 subcodes, we obtain the following results.

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$$|{\mathcal{C}_s^{\perp}}'| = 2^{2s}, \qquad |{\mathcal{C}_s^{\perp}}^{o}| = 2^{2s}.$$

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- **GC-weight enumerator of ${\mathcal{C}_s^{\perp}}'$ and ${\mathcal{C}_s^{\perp}}^{o}$**

$$GCW_{{\mathcal{C}_s^{\perp}}'}(x, y) = GCW_{{\mathcal{C}_s^{\perp}}^{o}}(x, y) = 2^s x^{2^s} + (2^{2s} - 2^{s+1}) x^{2^{s-1}} y^{2^{s-1}} + 2^s y^{2^s}.$$

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- **GC-weight enumerator of $\mathcal{C}_s^{\perp}$**

$$GCW_{\mathcal{C}_s^{\perp}}(x, y) = 2^{s+1} x^{2^s} + 2^{s+2}(2^s - 1)x^{2^{s-1}} y^{2^{s-1}} + 2^{s+1} y^{2^s}.$$

Main Results

From the analysis of all 64 subcodes, we obtain the following results.

- **Self-Reversible & Self Reverse-Complement Codewords**

$$|\mathcal{C}_s^{\perp'}| = 2^{2s}, \qquad |\mathcal{C}_s^{\perp^o}| = 2^{2s}.$$

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$$GCW_{\tilde{\mathcal{C}_s^{\perp}}''}(x,y) = GCW_{\tilde{\mathcal{C}_s^{\perp}}^{oo}}(x,y) = 2^{s-1} x^{2^s} + 2^s(3\cdot 2^{s-1} - 1) x^{2^{s-1}} y^{2^{s-1}} + 2^{s-1} y^{2^s}.$$

Table: Families of DNA codes constructed from $\mathcal{C}_s^{\perp}$

Family	Constraints satisfied	Size
$\mathcal{C}_s^{\tilde{\perp}''}$	Reverse	$3 \cdot 2^{2s-1}$
$\mathcal{C}_s^{\tilde{\perp}oo}$	Reverse-complement	$3 \cdot 2^{2s-1}$
$\mathcal{C}_{s,1}^{\perp}$	Fixed GC-content 2^{s-1}	$2^{s+2}(2^s - 1)$
$\mathcal{C}_s^{\tilde{\perp}''} \cap \mathcal{C}_{s,1}^{\perp}$	Reverse + fixed GC-content 2^{s-1}	2^{s-1}
$\mathcal{C}_s^{\tilde{\perp}oo} \cap \mathcal{C}_{s,1}^{\perp}$	Reverse-complement + fixed GC-content 2^{s-1}	$2^s(3 \cdot 2^{s-1} - 1)$

In each case, the Hamming distance constraint is satisfied with distance at least 2^{s-1}.

DNA codes from $\mathcal{C}_s$

- Define the reverse map $T : \mathcal{C}_s \to \mathcal{C}_s$, by
$$T(c_0, c_1, \ldots, c_{2^s-1}) = (c_{2^s-1}, \ldots, c_1, c_0).$$

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- The set of self-reversible codewords

$$\mathcal{C}'_s = \ker(T - I).$$

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$$\mathcal{C}_s'' = \mathcal{C}_s \setminus \mathcal{C}_s'.$$

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$$\mathcal{C}_s'' = \tilde{\mathcal{C}_s''} \cup \tilde{\mathcal{C}_s''}^R, \quad \text{where } \tilde{\mathcal{C}_s''}^R = \{c^R \mid c \in \tilde{\mathcal{C}_s''}\}.$$

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- Therefore,
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Result

$\tilde{\mathcal{C}_s''}$ produces DNA codes satisfying R-constraint with Hamming distance at least 4.

- The GC-weight enumerator of $\mathcal{C}_s$ is

$$GCW_{\mathcal{C}_s}(x,y) = 2^{2^s-2s-2}\left[(x+y)^{2^s} + (x-y)^{2^s} + (2^{s+1}-2)(x+y)^{2^{s-1}}(x-y)^{2^{s-1}}\right].$$

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- If the GC-weight enumerator of the self-reversible part $\mathcal{C}_s'$ is known, then

$$GCW_{\mathcal{C}_s''}(x,y) = GCW_{\mathcal{C}_s}(x,y) - GCW_{\mathcal{C}_s'}(x,y).$$

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Construction of DNA Codes from $\mathcal{C}_s$

- From the GC-weight enumerator of $\mathcal{C}_s$, we obtain DNA codes satisfying fixed GC-content.
- From the GC-weight enumerator of $\tilde{\mathcal{C}_s''}$, we obtain DNA codes satisfying both fixed GC-content and the R-constraint.

In both cases, the Hamming distance constraint is satisfied with distance at least 4.

DNA codes from $\mathcal{C}_4$

- The GC-weight enumerator of $\mathcal{C}_4$

$$2048x^{16}+286720x^{12}y^{4}+917504x^{10}y^{6}+1781760x^{8}y^{8}+917504x^{6}y^{10}+286720x^{4}y^{12}+2048y^{16}.$$

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$$|\mathcal{C}_4'| = |\ker(T-I)| = 4^7 = 16384.$$

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- The GC-weight enumerator of the set of self-reversible codewords

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- Thus, the GC-weight enumerator of $\tilde{\mathcal{C}_4''}$
  $960x^{16} + 141568x^{12}y^4 + 458752x^{10}y^6 + 886400x^8y^8 + 458752x^6y^{10} + 141568x^4y^{12} + 960y^{16}.$

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DNA Codes Obtained from $\mathcal{C}_4$

We obtain DNA codes of length 16 and minimum Hamming distance at least 4:

Constraint	GC_6	GC_8	GC_{10}	R	$R+GC_6$	$R+GC_8$	$R+GC_{10}$
Size	917504	1781760	917504	2088960	458752	886400	458752

m-Secondary Structure Avoiding DNA Sequences

Definition

A DNA sequence c of length n is called an m-secondary structure avoiding sequence (m-SSA) if there do not exist overlapping subsequences $u, v \subseteq c$ such that

$$v = u^{RC} \quad \text{and} \quad |u| = |v| \geq m.$$

$$c = (c_1, \ldots, \underbrace{c_i, c_{i+1}, \ldots, c_{i+m-1}}_{u}, \ldots, \underbrace{c_j, c_{j+1}, \ldots, c_{j+m-1}}_{v}, \ldots, c_n),$$

where u and v are overlapping subsequences of c satisfying $v = u^{RC}$.

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Example

The sequence

$$\underbrace{\mathtt{ATAG}}_{u}\,\mathtt{CAG}\,\underbrace{\mathtt{CTAT}}_{v=u^{RC}}$$

contains complementary overlapping segments and may form a hairpin structure.

A DNA code $\mathcal{C}$ is called an m-SSA code if every codeword in $\mathcal{C}$ is an m-SSA sequence.

m-SSA DNA Codes from $\mathcal{C}_s$ and $\mathcal{C}_s^{\perp}$

Code	Parameters	m	# of m-SSA codewords
$\mathcal{C}_3^{\perp}$	$[8, 4, 4]$	$2, 3, 4$	$92, 168, 192$
$\mathcal{C}_4^{\perp}$	$[16, 5, 8]$	$2, 3, 4, 5$	$132, 252, 356, 660$
$\mathcal{C}_5^{\perp}$	$[32, 6, 16]$	$2, 3, 4, 5, 7, 8, 9$	$260, 316, 372, 900, 1308, 1412, 2628$
$\mathcal{C}_6^{\perp}$	$[64, 7, 32]$	$2, 5, 7, 9, 13, 17$	$516, 964, 1276, 3492, 5124, 10500$
$\mathcal{C}_7^{\perp}$	$[128, 8, 64]$	$7, 9, 13, 15, 17$	$2300, 3556, 4804, 5116, 13860$

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$\mathcal{C}_6^\perp$	$[64,7,32]$	$2,5,7,9,13,17$	$516,964,1276,3492,5124,10500$
$\mathcal{C}_7^\perp$	$[128,8,64]$	$7,9,13,15,17$	$2300,3556,4804,5116,13860$

Codes from $\mathcal{C}_s$

For $s = 4$, the code $\mathcal{C}_4 = [16, 11, 4]$ produces

m	# of m-SSA codewords
2	60676
3	1969808
4	3444804
5	4083668

Conclusion & Future Directions

Conclusion

- Established reverse and reverse-complement closure properties of the linear codes $\mathcal{C}_s$ and $\mathcal{C}_s^{\perp}$ over $\mathbb{F}_4$.
- Determined the GC-weight enumerator of these codes and their subcodes containing self-reversible and self-reverse-complement codewords.
- Obtained DNA codes satisfying
  - R-constraint
  - RC-constraint,
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Future Directions

- Construction of codes over $\mathbb{F}_4$ and derivation of DNA codes with biological constraints.
- Development of DNA codes with secondary structure avoidance.
- Study of DNA codes correcting insertion and deletion errors.
- Applications to DNA storage and DNA computing.

Thank You for the Attention!